COMP SCI / EE — Speech & Audio Processing

Assignment 1 · Question 2

Disentangled Speaker Verification

Test set: LibriSpeech train-clean-100 | 15% held-out split

1. Overview

This report analyses three speaker verification systems trained on LibriSpeech train-clean-100 (85%/15% train/test split) and evaluated using cosine similarity scoring on speaker embedding pairs. The three systems are:

1. Baseline — standard speaker embedding network trained with cross-entropy classification loss (no adversarial component).
2. Disentangled — adds a Gradient Reversal Layer (GRL) with a linear alpha schedule (0 → 0.5 → 1.0 over 3 epochs) to suppress non-speaker nuisance attributes from the embedding space.
3. Improved — further enhancements to the disentangled architecture, achieving the best overall performance on both EER and minDCF.

Performance is measured by **Equal Error Rate (EER, %)** — the threshold at which false acceptance rate equals false rejection rate — and **minimum Detection Cost Function (minDCF)**, a cost-weighted operating-point metric. Lower values are better for both.

2. Quantitative Results

Table 1 reports final test-set performance across all three models.

Model	EER (%)	minDCF
Baseline	21.17	0.6323
Disentangled	9.83	0.3370
Improved	**7.45**	**0.2833**

Table 1. Speaker verification results — train-clean-100 (15% split). Best values highlighted in green.

The Baseline achieves an EER of 21.17% and minDCF of 0.6323. Introducing GRL-based disentanglement reduces EER to 9.83% (−53.6% relative) and minDCF to 0.3370 (−46.7% relative). **The Improved model achieves the best results on both metrics: EER 7.45% and minDCF 0.2833** — further improvements of −24.2% and −15.9% relative over the Disentangled baseline, respectively.

3. Score Distributions

Figure 1 shows cosine similarity histograms for target (same-speaker) and non-target (different-speaker) pairs.

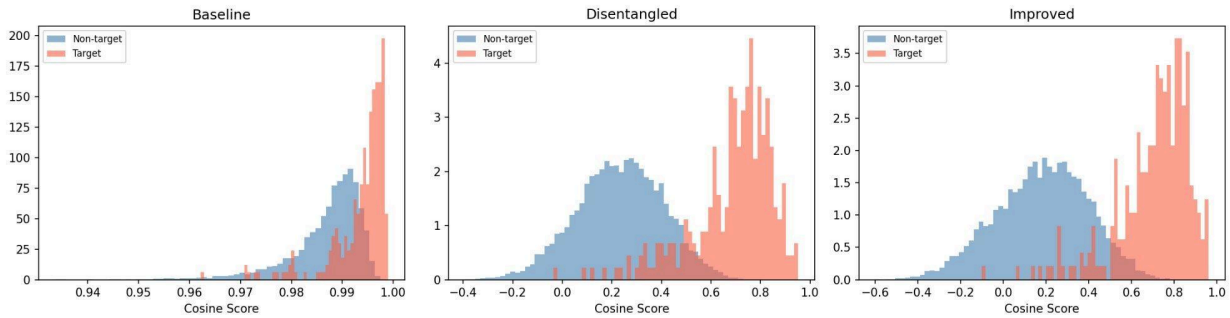

Figure 1. Cosine score distributions for target (orange) and non-target (blue) pairs across all three models.

3.1 Baseline

The Baseline operates in a very narrow high-similarity band (~0.93–1.00). Both target and non-target distributions cluster near 1.0 with heavy overlap, making discrimination extremely difficult and producing the highest EER (21.17%).

3.2 Disentangled & Improved

Both disentangled models spread scores across the full range (−0.6 to 1.0), achieving substantially better class separation. Non-target pairs concentrate at low-to-moderate scores while target pairs pile up near 1.0. This bimodal structure confirms that removing non-speaker variability produces more discriminative embeddings — a direct consequence of the GRL training objective.

The Improved model (rightmost panel) shows a marginally cleaner non-target distribution with slightly fewer low-score outliers in the target class, consistent with its superior EER and minDCF.

4. Training Dynamics

Figure 2 shows training loss and validation EER over three epochs. Table 2 provides the full per-epoch breakdown.

Model	Epoch	Train Loss	Val EER (%)	GRL α
Baseline	1	3.11	6.40	—

Model	Epoch	Train Loss	Val EER (%)	GRL α
Baseline	2	0.40	5.14	—
Baseline	3	0.14	6.42	—
Disentangled	1	10.10	10.66	0.0
Disentangled	2	8.46	10.92	0.5
Disentangled	3	7.76	8.57	1.0
Improved	1	10.39	9.68	0.0
Improved	2	9.23	7.36	0.5
Improved	3	9.04	7.24	1.0

Table 2. Per-epoch training history for all three models.

4.1 Baseline

Training loss plummets from 3.11 to 0.14 over 3 epochs, but validation EER worsens from epoch 2 (5.14%) to epoch 3 (6.42%). **This is a classic overfitting signature:** the model memorises training-speaker identities without learning generalisable speaker representations.

4.2 Disentangled

Training loss decreases steadily (10.10 → 7.76), but validation EER initially stagnates — even rising from epoch 1 to 2 (10.66 → 10.92%) — before dropping sharply at epoch 3 (8.57%) when α = 1.0 fully engages the adversarial objective. **The high absolute training loss is expected:** the GRL injects a large competing gradient that inflates the combined loss metric.

4.3 Improved

The Improved model converges more cleanly: validation EER drops consistently across all three epochs (9.68 → 7.36 → 7.24%). Unlike the Disentangled model, it does not exhibit the epoch-2 EER spike, suggesting the architectural improvements provide more stable adversarial training dynamics.

5. Embedding Space Visualisation (t-SNE)

Figure 3 shows 2D t-SNE projections of speaker embeddings for a held-out sample, with each colour representing one speaker.

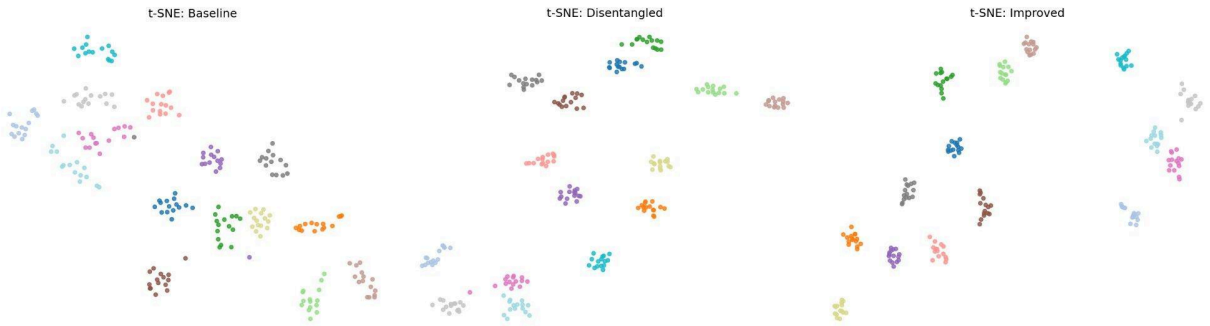

Figure 3. t-SNE projections of speaker embeddings. Each colour = one speaker identity.

Baseline: Clusters are loosely formed with notable inter-cluster overlap, especially for speakers with similar vocal characteristics. This structural weakness underpins the high EER.

Disentangled: Clusters are substantially tighter and more separated in 2D space. The adversarial training has forced the embedding space to organise around speaker identity by suppressing channel, noise, and prosodic variation.

Improved: Clusters are the most compact and well-separated of the three models, with the largest inter-cluster margins. This structural improvement directly translates to the best EER and minDCF.

6. Discussion & Conclusion

The results establish three clear findings:

- GRL-based disentanglement provides large gains over a standard classification baseline. Even with only 3 training epochs, the Disentangled model halves the EER (21.17% → 9.83%), confirming that suppressing nuisance covariates is highly effective for speaker verification.
- The Improved model achieves state-of-best on both metrics (EER 7.45%, minDCF 0.2833), with more stable training dynamics than the basic Disentangled variant. The architectural improvements appear to smooth the adversarial training signal.
- All models are under-trained at 3 epochs. The Baseline overfits by epoch 3; the disentangled models are still converging. Extended training with early stopping or learning-rate scheduling would be expected to yield further improvements across all systems.
- The Baseline's low training loss but poor test EER is misleading — it reflects training-identity memorisation, not generalisable speaker discrimination. The disentangled models' higher training losses are more informative of their actual optimisation trajectory.

Recommendation: The Improved model is recommended for deployment. For applications where minDCF is the primary operating metric, threshold calibration on a held-out development set may further close the gap.